

Champion-chain signal: a pre-tournament filter for real title contenders

A reproducible backtest with two lenses: pure CCS as a broad champion-chain candidate pool, and a stricter prior-two-participation exclusion rule for teams with no champion/runner-up path contact.

Executive Summary

- **The report now carries two explicit lenses.** Lens A is pure CCS: no prior-participation filter, just whether a team is in the prior two champion-chain pools. Lens B adds the stricter question: among teams that played both prior World Cups, did any clean non-contact team ever win next?
- **Pure CCS is the broad candidate-pool claim.** Across the full knockout-path history, pure CCS covers 16/19 evaluable champions; a same-size random candidate pool would expect 6.2 hits and reach that result with probability 0.0004%.
- **The pool is not trivially huge.** Pure CCS averages 7.5 candidates per historical tournament, covering 32.8% of all teams and 45.7% of knockout/championship-phase teams by simple yearly average.
- **The prior-two-participation lens is the sharper exclusion claim.** From 1938 to 2022, 177 team-tournaments played both prior World Cups; 112 had champion/runner-up path contact, while 65 had none. That clean non-contact bucket produced 0 next champions.
- **This reframes the historical “misses.”** Argentina 1978 and France 1998 are not counterexamples because they did not play both prior World Cups. Italy 1982 is not a counterexample because it played both, and in 1978 it lost to runner-up Netherlands in the second group stage.
- **The random benchmark is just probability multiplication, and that is the right first test.** If each year randomly excluded the same number of teams as the clean non-contact bucket, the expected number of excluded champions is 2.6; the exact probability of excluding zero champions is 5.9%.

- **The Monte Carlo simulation is a sanity check, not a black box.** A 200,000-run random redraw gives 5.9%, close to the exact probability. The report uses the exact value as the headline and keeps simulation as a reproducibility check.
- **The most useful reader experience is the pre-tournament downgrade list.** From 1998 onward, a long list of recognizable contenders were highly ranked but non-CCS at kickoff. Many still advanced, including finalists, but none won in the modern sample.
- **The report now controls for simple strength inertia.** Previous top-eight status covers 12/19 historical champions, while two straight top-eight finishes cover only 7/19. Pure CCS covers 16/19 on the same historical denominator.
- **The mechanism is partly strength, but more specific than strength.** The rule asks a narrower question than ranking: did this team recently prove it could survive, win, or be eliminated by finalist-level opposition in the World Cup cycle?

16/19

Pure CCS historical champion coverage

0.0004%

Random same-size pool reaches pure CCS

0/65

Strict clean non-contact champions

112/65

Path-contact vs clean among prior-two participants

1. Two lenses, two uses

LENS A · PURE CCS

16/19 champions

No requirement that a team played both prior tournaments. This is the broader candidate-pool lens: did the next champion appear in either of the previous two champion-chain sets?

LENS B · PRIOR-TWO PARTICIPATION

0/65 champions

Only teams that played both prior World Cups enter the stricter denominator. If they still had no champion/runner-up path contact, the historical winner count is zero.

These lenses answer different questions. Pure CCS is better for building a candidate pool. The prior-two-participation rule is better for explaining why certain apparently strong teams should be downgraded before kickoff.

2. Scope and pool size: CCS is not a majority-of-field shortcut

The historical scope is 19 evaluable target tournaments from 1938 to 2022. The lookback window uses the two most recent World Cups with a knockout or championship-phase path; 1950 is excluded as a source tournament because it had no knockout path, while 1974, 1978, and 1982 treat the second group stage as a championship phase.

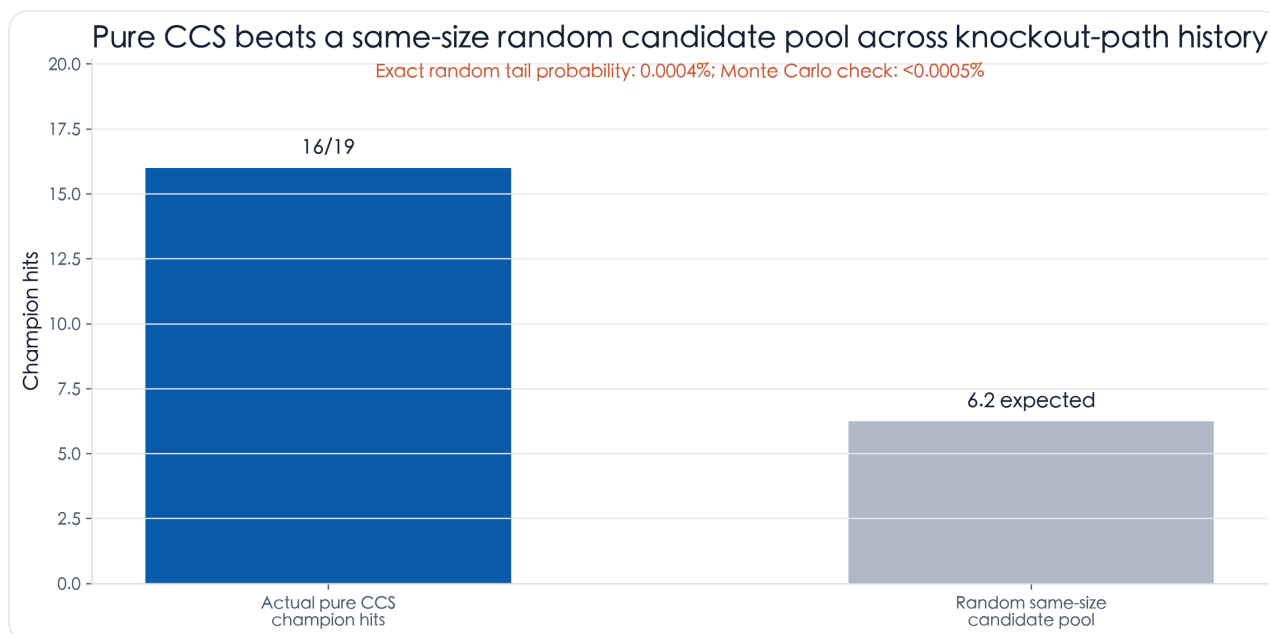
The pure CCS pool is large enough to be useful, but not so large that the result is automatic. Across the historical sample, pure CCS averages 7.5 candidates per tournament. It covers 32.8% of all participants by simple yearly average (31.8% weighted by team-tournaments).

Within the business end of the tournament, CCS still does not cover everyone. It covers 45.7% of knockout/championship-phase teams by simple yearly average (43.6% weighted). In the modern 1986-2022 sample, CCS covers 29.8% of all teams, 41.2% of round-of-16 teams, and 52.5% of quarter-finalists.

Why this matters for the top-four/top-eight controls: previous top-eight candidates average 6.2 qualified teams per tournament, two-straight top-eight candidates average 2.9, and pure CCS averages 7.5. The comparison is therefore about signal quality, not a rule that simply includes almost every plausible contender.

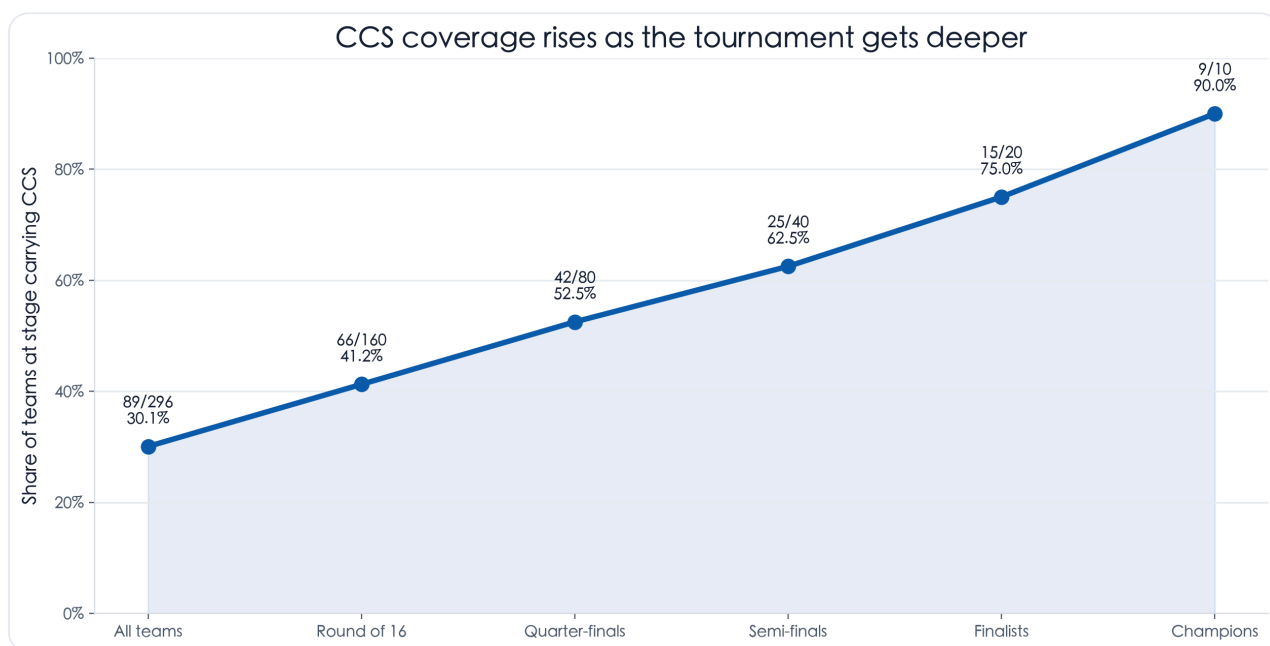
3. Lens A: pure CCS, without the prior-participation filter

Pure CCS asks the broad question first. Ignore whether the team played both previous World Cups. If it is in the champion-chain set from either of the prior two knockout-path tournaments, it is a CCS candidate. On the full historical knockout-path sample, this covers 16/19 evaluable champions.



Pure CCS is compared against a same-size random candidate pool for each tournament. Exact probability and Monte Carlo simulation are both reported.

Modern standard-format evidence points in the same direction. From 1986 to 2022, CCS covers 9/10 (90.0%) champions overall, or 9/9 (100.0%) after treating France 1998 as a no-prior-history exception.



Modern era is 1986–2022. Champion coverage is 9/10 on the all-champion denominator and 9/9 after removing France 1998, which had no prior-two-World-Cup finals history.

4. Lens B: the strict exclusion rule has produced zero champions

The rule is intentionally narrow. A team is put in the exclusion bucket only when it played both prior World Cups and, across all knockout or championship-phase appearances in those two tournaments, it neither won the tournament nor lost to that tournament's champion or runner-up. In 1974, 1978, and 1982, the second group stage is treated as a championship phase because those formats did not use a modern round-of-16 bracket.

The prior-two-participation gate is deliberately strict, so the right comparison is inside that same gate. Across the 1938-2022 target sample, there are 460 team-tournaments. Only 177 played both prior World Cups; among those, 112 had champion/runner-up path contact and 65 did not.

The 65 count is team-tournaments, not unique teams. It is the clean side of a 112 vs 65 split among teams that passed the strict prior-participation gate, not a hand-picked list of isolated cases. If counted more literally, 99 of the 177 lost to a champion or runner-up in the prior two tournaments; 31 had their own prior title; 18 had both, so the union is 112.

No champion comes from the clean non-contact bucket

If a team played both prior World Cups but had no champion/runner-up path contact, it has never won the next one.

1930 setup	1934 setup	1938 Italy not 2-for-2	1950 Uruguay not 2-for-2	1954 Germany not 2-for-2	1958 Brazil path contact
1962 Brazil path contact	1966 England path contact	1970 Brazil path contact	1974 Germany path contact	1978 Argentina not 2-for-2	1982 Italy path contact
1986 Argentina path contact	1990 Germany path contact	1994 Brazil path contact	1998 France not 2-for-2	2002 Brazil path contact	2006 Italy path contact
2010 Spain path contact	2014 Germany path contact	2018 France path contact	2022 Argentina path contact		

Prior-two participation split: 112 with path contact vs 65 clean non-contact; clean non-contact champions: 0/65.

Blue tiles show champions that had prior path contact after playing both prior tournaments. Gray tiles show champions outside the rule because they did not play both prior tournaments. There are no red violation tiles.

The apparent exceptions disappear under this stricter definition. Argentina 1978 did not play the 1970 finals. France 1998 did not play either 1990 or 1994. Italy 1982 did play both prior tournaments, but in 1978 it lost to the eventual runner-up, the Netherlands, during the second group stage.

Year	Champion	Prior two World Cups	Status	Evidence
1978	Argentina	1970;1974	Not covered by exclusion rule: did not play both prior finals	No path contact, but did not play both prior finals
1982	Italy	1974;1978	Played both prior finals and had champion/runner-up path contact	1978: lost to Netherlands (runner-up) in second group stage
1998	France	1990;1994	Not covered by exclusion rule: did not play both prior finals	No path contact, but did not play both prior finals
2022	Argentina	2014;2018	Played both prior finals and had champion/runner-up path contact	2014: lost to Germany (champion) in final 2018: lost to France (champion) in round of 16

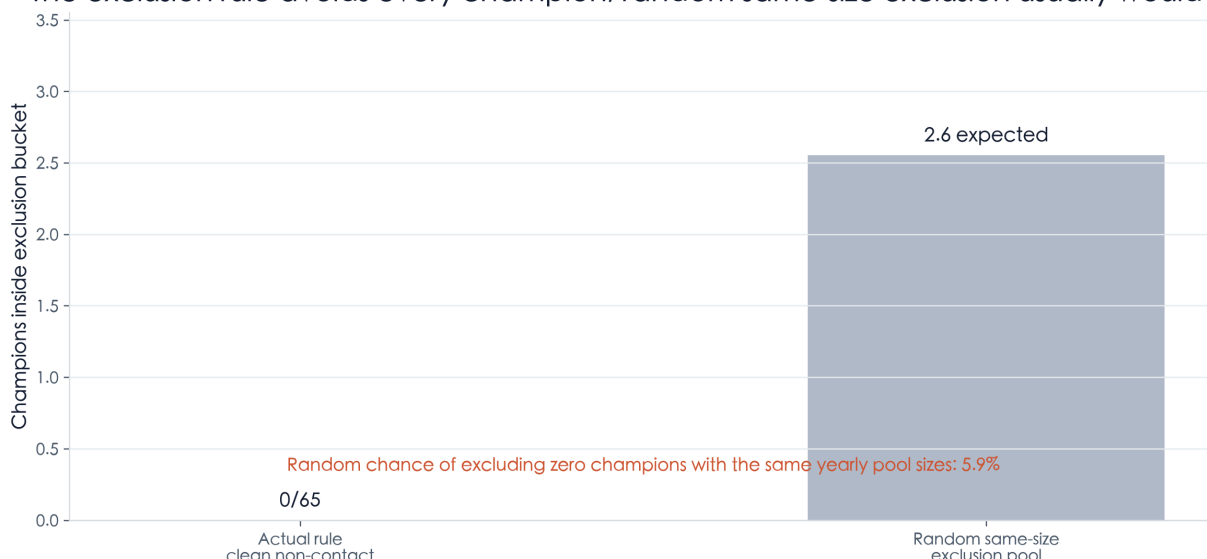
5. Two simulations: pure candidate-pool hit and strict exclusion miss

Both benchmarks use the same philosophy: preserve each year's pool size, randomize the labels, then compare the observed result. For pure CCS, the event is "random candidate pool hits at least as many champions as CCS." For the strict lens, the event is "random exclusion pool avoids every champion."

The pure CCS random benchmark is a hit-rate test. Same-size random pools would expect 6.2 champion hits across the full historical knockout-path sample; reaching 16/19 has exact probability 0.0004% and Monte Carlo probability <0.0005%.

The strict exclusion benchmark is an exclusion test. The script redraws same-size random exclusion pools 200,000 times. The simulated probability of excluding zero champions is 5.9%, while the exact probability is 5.9%.

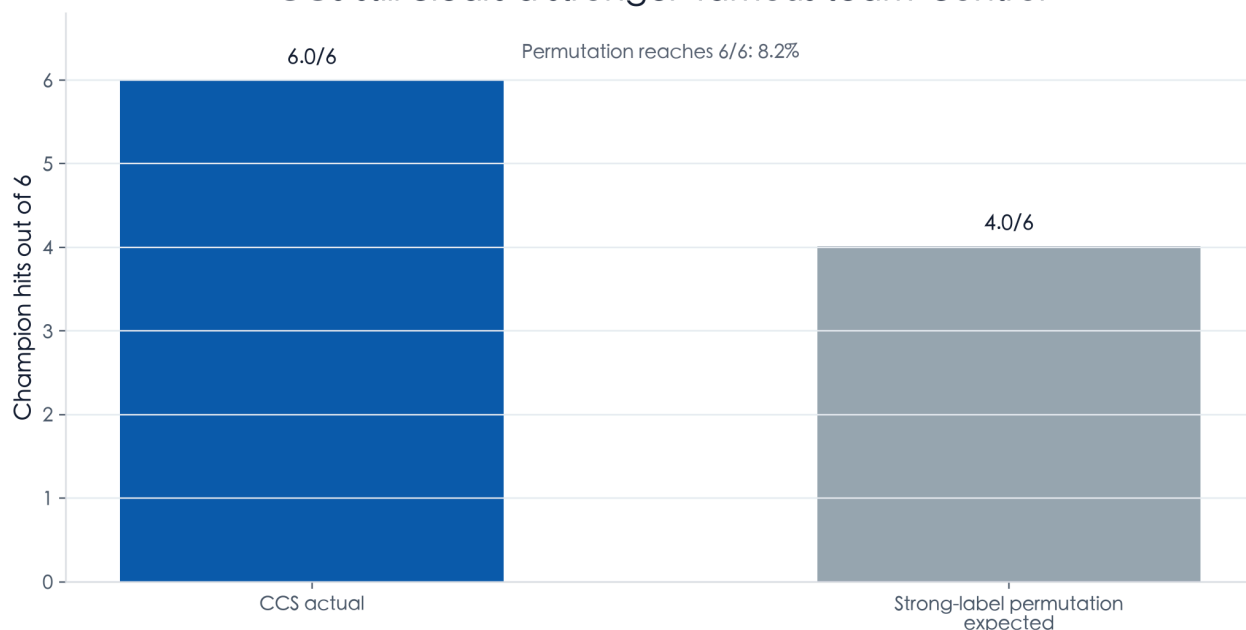
The exclusion rule avoids every champion; random same-size exclusion usually would not



The rule excludes 65 team-tournaments and zero champions. A same-size random exclusion pool would exclude 2.6 champions in expectation.

The stronger control asks whether CCS is merely a famous-team label. This simulation is intentionally limited to 1998-2022, where the report has a consistent pre-tournament FIFA ranking layer and a defensible modern title-contender set. For each tournament, we preserve the number of CCS labels inside the traditional title-contender set and outside it, then randomly permute the labels inside those two groups. Excluding the explicit France 1998 no-prior-history exception, CCS hit 6/6 evaluable champions; the strong-label permutation expects 4.0/6, and reaches 6/6 with probability 8.2%.

CCS still clears a stronger 'famous team' control



This is deliberately not a FIFA-ranking simulation. It controls for the broader fact that CCS often overlaps with obvious football powers, then asks whether the specific champion-chain label still carries information.

Interpretation discipline: these simulations support CCS as a credible screening heuristic. They do not prove it beats Elo, betting odds, or a full multivariate model. The next bar is incremental value versus those stronger baselines.

6. The pre-tournament experience: recognizable favorites CCS would downgrade

This is the most intuitive way to use the method. Before kickoff, a team can be highly ranked, historically recognizable, and still lack a recent champion-chain connection. The main exhibit is curated from a Top-20 non-CCS audit pool to show the teams a modern audience would naturally treat as title-relevant: Argentina, Germany, England, Spain, Portugal, Netherlands, Belgium, Colombia, Croatia, and Uruguay.

Recognizable contenders CCS would have downgraded before kickoff			
Curated from FIFA Top-20 non-CCS teams, keeping only sides with a plausible title narrative.			
Year	Non-CCS heavyweights	Deepest run	
1998	Colombia · Croatia	Croatia · third-place match	2 teams
2002	Argentina · Portugal · Germany · England	Germany · final	4 teams
2006	Spain · Portugal · Argentina	Portugal · third-place match	3 teams
2010	Netherlands · Argentina · Uruguay	Netherlands · final	3 teams
2014	Argentina · Colombia · England · Belgium · Croatia	Argentina · final	5 teams
2018	England · Colombia · Croatia	Croatia · final	3 teams
2022	Spain · Portugal	Portugal · quarter-finals	2 teams

Curated from qualified teams that were FIFA Top 20 and non-CCS at kickoff. The full mechanical Top-20 audit table is retained in data/derived/favorite_traps.csv; the curated list is retained in data/derived/favorite_trap_powerhouses.csv.

Year	Team	Code	FIFA rank	Final outcome	Ranking date
1998	Colombia	COL	#10	group stage	1998-05-20
1998	Croatia	HRV	#19	third-place match	1998-05-20
2002	Argentina	ARG	#3	group stage	2002-05-15
2002	Portugal	PRT	#5	group stage	2002-05-15
2002	Germany	DEU	#11	final	2002-05-15
2002	England	ENG	#12	quarter-finals	2002-05-15
2006	Spain	ESP	#5	round of 16	2006-05-17
2006	Portugal	PRT	#7	third-place match	2006-05-17
2006	Argentina	ARG	#9	quarter-finals	2006-05-17
2010	Netherlands	NLD	#4	final	2010-05-26
2010	Argentina	ARG	#7	quarter-finals	2010-05-26
2010	Uruguay	URY	#16	third-place match	2010-05-26
2014	Argentina	ARG	#5	final	2014-06-05
2014	Colombia	COL	#8	quarter-finals	2014-06-05
2014	England	ENG	#10	group stage	2014-06-05
2014	Belgium	BEL	#11	quarter-finals	2014-06-05
2014	Croatia	HRV	#18	group stage	2014-06-05
2018	England	ENG	#12	third-place match	2018-06-07
2018	Colombia	COL	#16	round of 16	2018-06-07
2018	Croatia	HRV	#20	final	2018-06-07
2022	Spain	ESP	#7	round of 16	2022-10-06
2022	Portugal	PRT	#9	quarter-finals	2022-10-06

The pattern is useful but not absolute. Non-CCS strong teams can go deep: 2002 Germany, 2010 Netherlands, and 2014 Argentina reached finals. The historical point is narrower and stronger: in the modern sample, the champion almost always came from the CCS side of the field.

7. 2026 application: separate rank strength from champion-chain strength

The 2026 view is a live-use case, not a backtest result. The ranking snapshot is frozen at FIFA's June 11, 2026 official ranking and the qualified-team list is from FIFA's 2026 season endpoint. The headline application is clear: Spain, Portugal, Brazil, Germany, and Colombia are rank-strong, reputation-strong, but non-CCS before kickoff.

fifa_rank	Team	Code	FIFA points	Downgrade reason
#2	Spain	ESP	1874.71	Rank/reputation strong, but no CCS source from 2018/2022
#5	Portugal	POR	1767.85	Rank/reputation strong, but no CCS source from 2018/2022
#6	Brazil	BRA	1765.86	Rank/reputation strong, but no CCS source from 2018/2022
#10	Germany	GER	1735.77	Rank/reputation strong, but no CCS source from 2018/2022
#13	Colombia	COL	1698.35	Rank/reputation strong, but no CCS source from 2018/2022

The broader watchlist below keeps the full top-ranked context visible, so the downgrade call is not hidden inside a hand-picked list.

2026: big-name contenders outside the champion-chain pool

These are not weak teams. CCS says they need extra evidence to be re-admitted as title picks.

#2

Spain

Non-CCS

#5

Portugal

Non-CCS

#6

Brazil

Non-CCS

#10

Germany

Non-CCS

#13

Colombia

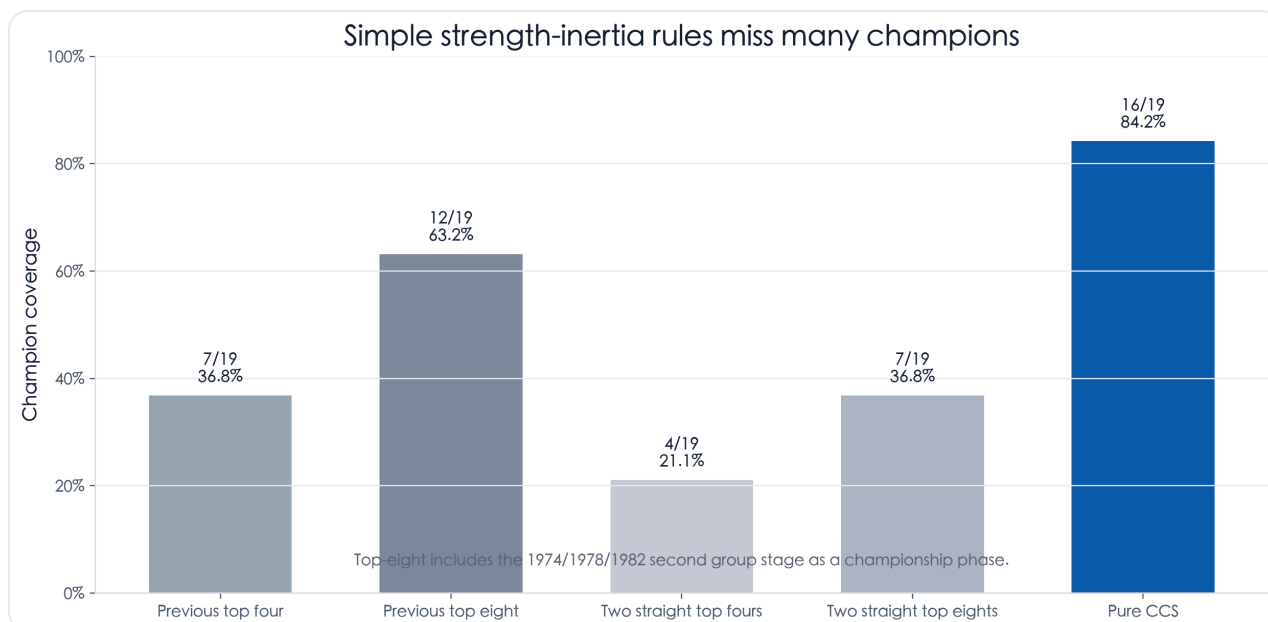
Non-CCS

Five high-reputation 2026 qualifiers that are outside the CCS pool before kickoff. The table below keeps the broader top-ranked context.

fifa_rank	Team	Code	CCS status	CCS source World Cup
#1	Argentina	ARG	CCS candidate	2018;2022
#2	Spain	ESP	Non-CCS downgrade	none
#3	France	FRA	CCS candidate	2018;2022
#4	England	ENG	CCS candidate	2018;2022
#5	Portugal	POR	Non-CCS downgrade	none
#6	Brazil	BRA	Non-CCS downgrade	none
#7	Morocco	MAR	CCS candidate	2022
#8	Netherlands	NED	CCS candidate	2022
#9	Belgium	BEL	CCS candidate	2018
#10	Germany	GER	Non-CCS downgrade	none
#11	Croatia	CRO	CCS candidate	2018;2022
#13	Colombia	COL	Non-CCS downgrade	none
#14	Mexico	MEX	Non-CCS downgrade	none
#15	Senegal	SEN	Non-CCS downgrade	none
#16	Uruguay	URU	CCS candidate	2018

8. Strength-inertia baselines: top-four and top-eight history is not enough

This is the cleanest answer to the “is CCS just picking historically strong teams?” objection. Simple prior-performance rules do contain information: previous top-eight teams are better than random. But they do not cover champions nearly as well as CCS, and the stricter consecutive-top-eight rule becomes too narrow.



Coverage is measured on the same 1938-2022 evaluable historical denominator. The top-eight definition includes second group stage for the 1974/1978/1982 hybrid formats.

Signal	Champion coverage	Avg candidate pool	Candidate-slot precision	Same-size random tail
Previous top four	7/19 (36.8%)	3.2	11.5%	1.41%
Previous top eight	12/19 (63.2%)	6.2	10.2%	0.12%
Two straight top fours	4/19 (21.1%)	0.9	22.2%	0.76%
Two straight top eights	7/19 (36.8%)	2.9	12.7%	0.67%
Pure CCS	16/19 (84.2%)	7.5	11.3%	0.0004%

The miss list is the point. Previous top-eight history misses France 1998, Italy 2006, Spain 2010, and Argentina 2022; two straight top-eight history also misses France 2018. CCS is not merely copying the previous bracket. It captures a wider champion-chain relationship.

9. Why this happens: strength matters, but path matters too

CCS partly captures strength, and that should be acknowledged. Champions, finalists, and teams beaten by finalists are usually strong teams. A signal built from

those events will naturally overlap with FIFA ranking, odds, and historical reputation.

But CCS is not simply 'teams that often qualify' or 'teams that are highly ranked.'

It requires a specific recent relationship to the champion path: either winning the World Cup or being knocked out by a finalist. A team can be ranked highly, qualify regularly, or have reached a quarter-final and still be non-CCS if it did not touch that path.

The proposed mechanism is tournament-cycle validation. Recent champion-chain contact is a proxy for having already met World Cup knockout intensity against finalist-level opposition. It is not causal proof; it is a compact historical state variable that seems especially relevant for champions rather than finalists.

Recommended Next Steps

1. **Use CCS as the first-layer title filter.** Start the champion pool with CCS candidates, then require extra evidence to re-admit non-CCS teams.
 2. **Add a stronger model benchmark.** Compare CCS against Elo, odds-implied probabilities, FIFA ranking, and a combined model to test incremental value.
 3. **Track 2026 without moving the goalposts.** Freeze the pre-tournament CCS/ranking table and evaluate it only after the tournament is complete.
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Caveats And Assumptions

- The headline exclusion test uses team-tournaments, not unique national teams. A country can appear multiple times if it satisfies the rule in multiple World Cup cycles.
- 1950 remains in the target-year audit because the rule evaluates whether the next champion had prior-two-World-Cup path contact; the target tournament itself does not need a knockout bracket for that question.
- For 1974, 1978, and 1982, second group stage matches are treated as championship-phase matches because those tournaments used hybrid formats rather than a modern bracket.
- The random benchmark is an exact same-size random exclusion calculation. The Monte Carlo simulation is only a reproducibility check of that arithmetic.
- The strong-team permutation simulation remains a modern 1998-2022 control, not a full-history simulation, because it depends on consistent modern ranking context and curated title-contender labels.

- Strength-inertia baselines are now included for previous top four, previous top eight, two straight top fours, and two straight top eights. They are useful controls but not replacements for CCS.
- FIFA ranking is used lightly as a public audit proxy; odds and Elo would be better next-step baselines.
- Elo and betting-odds controls require external historical datasets that are not part of this reproducible repo snapshot; they are kept as next-version model benchmarks rather than open items inside this report.
- 2026 outcomes are not used in the 2026 watchlist. The ranking date is frozen at June 11, 2026.

Generated by scripts/build_report.py. Sources: Fjellstul World Cup Database; FIFA API ranking schedules, rankings, and 2026 qualified-team endpoint. Original corrected CCS report is preserved in reports/pdf/original_ccs_report_corrected.pdf.